THE INVARIANT

Genetic Drift — Formal Appendix

A rigorous treatment of the Markov chain structure,
the martingale property, and absorption

This appendix is a companion to the article *Genetic Drift* published on The Invariant. It assumes the reader has read the article. Notation and setup follow directly from the main text. Simulations verifying the results below are available at github.com/theinvariantlab/the-invariant.

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1. What kind of process is this?

Setup. Let $N \geq 1$ be the total population size (fixed). Let X_n denote the number of red shells at generation n. The state space is $\mathcal{S} = \{0, 1, \ldots, N\}$. States 0 and N are *absorbing*: once entered, the chain stays there. All interior states are *transient*.

Theorem 1.1 (Markov property). *The process $(X_n)_{n\geq 0}$ is a time-homogeneous Markov chain on $\mathcal{S}$, with transition probabilities*

$$P(X_{n+1} = k \mid X_n = i) = \binom{N}{k}\left(\frac{i}{N}\right)^k\left(1-\frac{i}{N}\right)^{N-k}, \quad k \in \mathcal{S}.$$

Proof. At each generation, every new shell independently inherits its color from a parent chosen uniformly at random from the current population. Given $X_n = i$, each new shell is red with probability i/N, independently of all others. Hence $X_{n+1} \mid X_n = i \sim \text{Binomial}(N, i/N)$, which gives the stated transition probabilities. These depend only on i, not on the history $X_0, \ldots, X_{n-1}$, so the chain is time-homogeneous.

In plain terms. To predict what the population looks like next generation, all you need is the current count. The entire history is represented by a single number.

2. Does either color have an advantage?

Theorem 2.1 (Martingale property). *The rescaled process $M_n = X_n/N$ is a martingale with respect to the natural filtration $(\mathcal{F}_n)_{n\geq 0}$:*

$$\mathbb{E}[M_{n+1} \mid \mathcal{F}_n] = M_n \quad \textit{a.s.}$$

Proof. Since $X_{n+1} \mid X_n \sim \text{Binomial}(N, X_n/N)$,

$$\mathbb{E}[X_{n+1} \mid \mathcal{F}_n] = N \cdot \frac{X_n}{N} = X_n.$$

Dividing by N gives the result.

In plain terms. The expected frequency of red tomorrow equals the frequency of red today. Neither color is being pushed toward winning at any step.

Since (M_n) is bounded in $[0, 1]$, it is bounded in L^2. By the martingale convergence theorem, $M_n \to M_\infty$ almost surely for some $[0, 1]$-valued random variable M_∞.

Corollary 2.2. *$M_\infty \in \{0,1\}$ almost surely. Equivalently, X_n reaches either 0 or N with probability 1.*

Proof. Since (M_n) is bounded in L^2, the increments are orthogonal and their squared norms are summable:

$$\sum_{k=1}^{\infty} \mathbb{E}[(M_k - M_{k-1})^2] < \infty.$$

See [3], Theorem 12.1. Since $\mathbb{E}[(M_k - M_{k-1})^2 \mid \mathcal{F}_{k-1}] = M_{k-1}(1 - M_{k-1})/N$, summing and taking expectations gives $\sum_k \mathbb{E}[M_{k-1}(1 - M_{k-1})] < \infty$. Because $M_n \to M_\infty$ a.s., if $M_\infty \in (0,1)$ on a set of positive probability then $M_n(1 - M_n)$ stays bounded away from zero on that set, making the sum diverge. This is a contradiction, so $M_\infty \in \{0,1\}$ a.s.

In plain terms. Even though neither color has an advantage at any given step, the random fluctuations accumulate over time. Eventually, one color disappears. The system is finite, and that is enough.

3. Who wins, and with what probability?

Theorem 3.1 (Fixation probability)**.** *Starting from $X_0 = i$, the probability that red fixes is*

$$\mathbb{P}(M_\infty = 1 \mid X_0 = i) = \frac{i}{N}.$$

Proof. Let $\tau = \inf\{n : X_n \in \{0, N\}\}$ be the absorption time. By Corollary 2.2, $\tau < \infty$ a.s. Applying the optional stopping theorem to the bounded martingale (M_n) and the stopping time τ,

$$\frac{i}{N} = M_0 = \mathbb{E}[M_\tau \mid X_0 = i] = 1 \cdot \mathbb{P}(X_\tau = N \mid X_0 = i) + 0 \cdot \mathbb{P}(X_\tau = 0 \mid X_0 = i).$$

Hence $\mathbb{P}(X_\tau = N \mid X_0 = i) = i/N$.

In plain terms. The starting count is the only thing that matters. How the population fluctuated in between is irrelevant. The initial frequency is the exact probability of winning.

Corollary 3.2 (Symmetry at equal split)**.** *If $X_0 = N/2$, then each color is equally likely to fix:*

$$\mathbb{P}(M_\infty = 1 \mid X_0 = N/2) = \mathbb{P}(M_\infty = 0 \mid X_0 = N/2) = \frac{1}{2}.$$

Proof. Immediate from Theorem 3.1 with $i = N/2$.

4. Does the underdog take longer to win?

The previous results say nothing about time, only about eventual outcomes. A natural follow-up: if a rare color manages to fix, does it take longer than a common one?

Result (Ewens [1], equations 3.5 and 3.11). The unconditional mean absorption time, starting from frequency $p = i/N$, satisfies the diffusion approximation

$$\mathbb{E}[\tau \mid X_0 = i] \approx -N\big[p \ln p + (1-p)\ln(1-p)\big].$$

This is the binary entropy function scaled by N. It peaks at $p = 1/2$ and vanishes at the boundaries: a population near fixation or near extinction absorbs quickly; one near equal split takes the longest on average.
The conditional mean fixation time, given that red actually wins, is longer for rare starting frequencies and shorter for common ones. A red population starting at $p = (2N)^{-1}$ that eventually fixes takes approximately $4N$ generations to do so. One starting at $p = 1/2$ takes approximately $2.8N$ generations.

The underdog, if it wins, wins slowly.

In plain terms. Conditioning on fixation selects the trajectories that traveled all the way from near 0 to 1. Those are long paths. The unconditional average is lower because most chains absorb quickly by losing.

In a finite system, randomness accumulates and eventually determines the outcome.

References

[1] W. J. Ewens, *Mathematical Population Genetics I: Theoretical Introduction*, 2nd ed., Springer, 2004.

[2] J. R. Norris, *Markov Chains*, Cambridge University Press, 1997.

[3] D. Williams, *Probability with Martingales*, Cambridge University Press, 1991.